DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Tutorial Sheet 5: Solutions **September 2024**

1. Let $V = \mathcal{C}([0,1],\mathbb{R})$ be the space of continuous functions from $[0,1]$ to $\mathbb{R}$. Set

$$N_1(f) = \int_0^1 x|f(x)|dx \quad \text{and} \quad N_2(f) := \int_0^1 x^2|f(x)|dx \qquad \forall f \in V\,.$$

   (a) Show that N_1 and N_2 are norms on V.

   (b) Show that the norms N_1 and N_2 are not equivalent. (Hint: Consider the sequence of functions $f_n(x) = \dfrac{1}{x^2 + \frac{1}{n}}$)

2. Let

$$\mathcal{C}_0 := \left\{x = \{x_n\} : \lim_{n\to\infty} x_n = 0\right\} \quad \text{equipped with norm} \quad \|x\|_\infty := \sup_{n\in\mathbb{N}} |x_n|.$$

   (a) Prove that $(\mathcal{C}_0, \|\cdot\|_\infty)$ is a Banach space.
   Let $(x^k)_k$ be a Cauchy sequence in $(\mathcal{C}_0, \|\cdot\|_\infty)$. That is,

$$\forall\varepsilon > 0 \quad \exists k_\varepsilon \in \mathbb{N} \text{ such that } k,\, h \geq k_\varepsilon \quad \Rightarrow \quad \left\|x^k - x^h\right\|_\infty = \max_{n\in\mathbb{N}} |x_n^k - x_n^h| < \varepsilon\,.$$

   So,

$$(*) \qquad\qquad k,\, h \geq k_\varepsilon \quad \Rightarrow \quad |x_n^k - x_n^h| < \varepsilon \quad \forall n \in \mathbb{N}\,.$$

   Thus for every $n \in \mathbb{R}$, $(x_n^k)_k$ is a Cauchy sequence in $\mathbb{R}$ and hence, converges to some real number x_n. Now, we set $x = (x_n)$ and we want to prove that $(x^k)_k$ converges in $(\mathcal{C}_0, \|\cdot\|_\infty)$ to x and $x \in \mathcal{C}_0$.
   Passing to the limit in $(*)$ when $h \to \infty$, we get that

$$k \geq k_\varepsilon \quad \Rightarrow \quad |x_n^k - x_n| < \varepsilon \quad \forall n \in \mathbb{N}$$

   which implies that

$$k \geq k_\varepsilon \quad \Rightarrow \quad \left\|x^k - x\right\|_\infty = \sup_{n\in\mathbb{N}} |x_n^k - x_n| \leq \varepsilon$$

   which shows that $(x^k)_k$ converges to x in the norm $\|\cdot\|_\infty$.
   Now taking $k = k_\varepsilon$, we get that

$$|x_n| \leq |x_n^{k_\varepsilon}| + |x_n^{k_\varepsilon} - x_n| < |x_n^{k_\varepsilon}| + \varepsilon \quad \forall n \in \mathbb{N}\,.$$

   Thus,

$$\limsup_{n\to\infty} |x_n| \leq \limsup_{n\to\infty} |x_n^{k_\varepsilon}| + \varepsilon = \lim_{n\to\infty} |x_n^{k_\varepsilon}| + \varepsilon = 0 + \varepsilon = \varepsilon$$

   and by the arbitrariness of $\varepsilon > 0$, we get that $\lim_{n\to\infty} x_n = 0$ and hence, $x = (x_n) \in \mathcal{C}_0$.

(b) Consider the mapping $L : \mathcal{C}_0 \to \mathbb{R}$ defined by

$$L(x) := \sum_{n=0}^{\infty} \frac{x_n}{n!} \qquad \forall x = \{x_n\} \in \mathcal{C}_0.$$

(i) Prove that $L \in \mathcal{C}_0'$ with $\|L\|_{\mathcal{C}_0'} = \sum_{n=0}^{\infty} \frac{1}{n!}$.

It is easy to see that L is a linear mapping. Notice that

$$|L(x)| \leq \sum_{n=0}^{\infty} \frac{|x_n|}{n!} \leq \|x\|_\infty \sum_{n=0}^{\infty} \frac{1}{n!}\,.$$

Hence, L is continuous and

$$\|L\|_{\mathcal{C}_0'} \leq \sum_{n=1}^{\infty} \frac{1}{n!}\,.$$

Let us prove that $\|L\|_{\mathcal{C}_0'} \geq \sum_{n=0}^{\infty} \frac{1}{n!}$. We just need to prove that $\|L\|_{\mathcal{C}_0'} \geq \sum_{n=1}^{k} \frac{1}{n!}$ for every $k \in \mathbb{N}$.

Let $k \in \mathbb{N}$. We set $e^k = (0, \dots, 0, 1, 0 \dots,)$ with the 1 at the kth position. Let

$$y^k := \sum_{n=0}^{k} e^n\,.$$

It is clear that $y^k \in c_0$ for every n and $\left\|e^k\right\|_\infty = 1$. Moreover,

$$\|L\|_{\mathcal{C}_0'} \geq L(y^k) = \sum_{n=0}^{\infty} \frac{y_n^k}{n!} = \sum_{n=0}^{k} \frac{1}{n!} \quad \forall k \in \mathbb{N}\,.$$

Thus, passing to the limit when $k \to \infty$ we find that

$$\|L\|_{\mathcal{C}_0'} \geq \sum_{n=0}^{\infty} \frac{1}{n!}\,.$$

(ii) Prove that this norm is not achieved in $\mathcal{C}_0$.

We want to show that there is no $x = (x_n)_n \in \mathcal{C}_0$ with $\|x\|_\infty = 1$ such

$$|L(x)| = \sum_{n=0}^{\infty} \frac{1}{n!}\,.$$

It is not restrictive to assume that $x_n \geq 0$ for every $n \in \mathbb{N}$. We get

$$\sum_{n=0}^{\infty} \frac{x_n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \quad \Leftrightarrow \quad \sum_{n=0}^{\infty} \frac{1 - x_n}{n!} = 0$$

$$\Leftrightarrow \quad \frac{1 - x_n}{n!} = 0 \quad \forall n \in \mathbb{N} \quad \text{(because } \|x\|_\infty = 1 \text{ and hence } x_n \leq 1)\,.$$

hence, $x_n = 1$ for every $n \in \mathbb{N}$. Thus, $x = (1, 1, \dots, \dots)$. for which $\lim_{n\to\infty} x_n = 1$ and therefore $x \notin \mathcal{C}_0$.

3. Let $\mathcal{C}^1([0,1])$ be the space of functions from $[0,1]$ to $\mathbb{R}$ which are continuous together with their first derivative. Set

$$H := \{f \in \mathcal{C}^1([0,1])\colon\ f(1/2) = 0\} \quad \text{and} \quad \sigma(f,g) := \int_0^1 e^x f'(x) g'(x) dx \quad \forall f,\, g \in H\,.$$

Prove that σ is a scalar product on H.

We want to show that the mapping σ satisfies the properties of a scalar product.

(i) $\forall f \in H$ $\sigma(f,f) = \displaystyle\int_0^1 e^x (f'(x))^2 dx \geq 0$ and

$$\begin{aligned} \sigma(f,f) = 0 \ &\Leftrightarrow \ \int_0^1 e^x (f'(x))^2 dx = 0 \ \Leftrightarrow \ e^x (f'(x))^2 = 0 \ \forall x \in [0,1] \\ &\Leftrightarrow \ f'(x) = 0 \ \forall x \in [0,1] \ \Leftrightarrow \ f \equiv C \text{ constant}\,. \end{aligned}$$

Since $f(1/2) = 0$, it follows that $C = 0$ and hence $f \equiv 0$.

(ii) $\forall f, g \in H$ $\sigma(f,g) = \displaystyle\int_0^1 e^x f'(x) g'(x) dx = \int_0^1 e^x g'(x) f'(x)) = \sigma(g,f)\,.$

(iii) $\forall f, g, h \in H, \ \ \forall \alpha, \beta \in \mathbb{R}$

$$\begin{aligned} \sigma(\alpha f + \beta g, h) &= \int_0^1 e^x (\alpha f'(x) + \beta g'(x)) h'(x) dx \\ &= \alpha \int_0^1 e^x f'(x) h'(x) dx + \beta \int_0^1 e^x g'(x) h'(x) dx \\ &= \alpha \sigma(f,h) + \beta \sigma(g,h)\,. \end{aligned}$$

Hence, the mapping σ is a scalar product.

4. Let $H = \mathbb{R}^3$ with the usual scalar product and $S := \operatorname{span}((1,3,0),\,(0,1,1))$ (the subspace spanned by the vectors $(1,3,0)$ and $(0,1,1)$) and $u = (1,1,0)$, then:

(a) Show that $u \notin S$;
By contradiction, if $u \in S$, there would exist $\alpha,\ \beta \in \mathbb{R}$ such that

$$u = (1,1,0) = \alpha(1,3,0) + \beta(0,1,1) = (\alpha, 3\alpha + \beta, \beta) \ \Leftrightarrow \ \begin{cases} \alpha = 1, \\ 3\alpha + \beta = 1 \\ \beta = 0 \end{cases}$$

which is impossible. So, $u \notin S$.

(b) Find the orthogonal projection $P_S(u)$ of u onto S.
Here from the definition of S and the definition of $P_S(u)$, we seek $\alpha, \beta \in \mathbb{R}$ such that

$$\begin{cases} P_S(u) = \alpha(1,3,0) + \beta(0,1,1) = (\alpha, 3\alpha + \beta, \beta), \\ ((1,1,0) - (\alpha, 3\alpha + \beta, \beta), (1,3,0)) = 0, \\ ((1,1,0) - (\alpha, 3\alpha + \beta, \beta), (0,1,1)) = 0. \end{cases} \quad \Rightarrow \quad \begin{cases} 10\alpha + 3\beta = 4 \\ 3\alpha + 2\beta = 1\,. \end{cases}$$

$$\Rightarrow \quad \alpha = \frac{5}{11} \text{ and } \beta = -\frac{2}{11}\,.$$

Thus,

$$P_S(u) = \frac{5}{11}(1,3,0) - \frac{2}{11}(0,1,1) = \left(\frac{5}{11}, \frac{13}{11}, -\frac{2}{11}\right)$$

(c) Deduce, $d_E(u,S)$ where d_E is the Euclidean distance on $\mathbb{R}^3$.
From the previous exercise, we find that

$$d_E(u,S) = \|u - P_S(u)\|_E = \left\|(1,1,0) - \left(\frac{5}{11}, \frac{13}{11}, -\frac{2}{11}\right)\right\|_E = \left\|\left(\frac{6}{11}, -\frac{2}{11}, \frac{2}{11}\right)\right\|_E = \frac{2}{\sqrt{11}}$$

5. Let $(V, \|\cdot\|)$ be a normed space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Show that the norm $\|\cdot\|$ satisfies the parallelogram identity if and only if

$$\|u+v\|^2 + \|u-v\|^2 \leq 2(\|u\|^2 + \|v\|^2) \qquad \forall u,\, v \in V\,. \qquad (*)$$

We just need to show that if $(*)$ holds then the parallelogram identity follows, that is

$$\|a+b\|^2 + \|a-b\|^2 = 2(\|a\|^2 + \|b\|^2) \qquad \forall a,\, b \in V,. \qquad (*)$$

So, let $a, b \in V$. Notice that, from $(*)$ we have

$$\|a+b\|^2 + \|a-b\|^2 \leq 2(\|a\|^2 + \|b\|^2\,.$$

Now, let us show that

$$\|a+b\|^2 + \|a-b\|^2 \geq 2(\|a\|^2 + \|b\|^2)\,. \qquad (**)$$

We set

$$u = \frac{a+b}{2} \quad \text{and} \quad v = \frac{a-b}{2} \quad \Rightarrow \quad \begin{cases} u+v = a, \\ u-v = b \end{cases}$$

and get from $(*)$

$$\begin{aligned}
\underbrace{\|u+v\|^2}_{=\|a\|^2} + \underbrace{\|u+v\|^2}_{=\|a\|^2} \leq 2(\|u\|^2 + \|v\|^2) \;&\Rightarrow\; \|a\|^2 + \|b\|^2 \leq 2\left[\frac{\|a+b\|^2}{4} + \frac{\|a-b\|^2}{4}\right] \\
&\Rightarrow\; \|a\|^2 + \|b\|^2 \leq \frac{1}{2}\left[\|a+b\|^2 + \|a-b\|^2\right] \\
&\Rightarrow\; 2(\|a\|^2 + \|b\|^2) \leq \|a+b\|^2 + \|a-b\|^2
\end{aligned}$$

which is (**).

6. Let $(H, (\cdot,\cdot))$ be a Hilbert space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let S, S_1, S_2 be subspaces of H. We recall that for every $A \subset H$, the orthogonal complement $A^\perp$ of A is defined as

$$A^\perp := \{u \in H\colon\; (u,v) = 0 \quad \forall v \in A\}\,.$$

(a) Let A, B be two subsets of H such that $A \subset B$. Show that $B^\perp \subset A^\perp$.

Let $v \in B^\perp$. Then $(u,v) = 0 \quad \forall u \in B$. In particular, for every $u \in A \subset B$, we get that $(u,v) = 0$. Hence, $v \in A^\perp$.

(b) Show that $S^\perp = \overline{S}^\perp$, where $\overline{S}$ denotes the closure of S in the normed space $(H, \|\cdot\|_H)$ with $\|\cdot\|_H$ being the norm induced by the scalar product $(\cdot,\cdot)$.

(c) Prove that $(S_1+S_2)^\perp = S_1^\perp \cap S_2^\perp$.

Double inclusion! $(S_1+S_2)^\perp \subset S_1^\perp \cap S_2^\perp$. In fact, $S_1 \subset S_1+S_2$ and $S_2 \subset S_1+S_2$ (Explain why!!) from which we obtain using (c) that

$$\begin{cases} (S_1+S_2)^\perp \subset S_1^\perp \\ (S_1+S_2)^\perp \subset S_2^\perp \end{cases} \Rightarrow \quad (S_1+S_2)^\perp \subset S_1^\perp \cap S_2^\perp\,.$$

On the other hand for the inclusion $S_1^\perp \cap S_2^\perp \subset (S_1+S_2)^\perp$, Let $u \in S_1^\perp \cap S_2^\perp$ and let $v \in S_1+S_2$. We want to show that $(u,v)=0$. Now, we can write $u = u_1+u_2$ with $u_1 \in S_1$ and $u_2 \in S_2$. Thus,

$$(u,v) = (u_1+u_2, v) = (u_1,v)+(u_2,v) = 0+0 = 0\,.$$

Therefore, $u \in (S_1+S_2)^\perp$.

(d) If S is a closed subspace, then show that $(S^\perp)^\perp = S$.

First of all, if $v \in S$, then $(v,w)=0$ for every $w \in S^\perp$. Hence, $v \in (S^\perp)^\perp$. Thus, $S \subset (S^\perp)^\perp$.
Now, let us show that $(S^\perp)^\perp \subset S$ when S is a closed subspace.For this, we make use of the orthogonal projection of onto S (this is where we use S closed subspace) as follows. Let $\boxed{w \in (S^\perp)^\perp}$. We write

$$w = \underbrace{P_S(w)}_{\in S} + \underbrace{(w-P_S(w))}_{\in S^\perp} \qquad \text{and we are going to show that } P_S(w)=w.$$

Notice that

$$\begin{cases} w \in (S^\perp)^\perp \\ w-P_S(w) \in S^\perp \end{cases} \Rightarrow \quad (w, w-P_S(w)) = 0\,. \quad (*)$$

On the other hand,

$$\begin{cases} P_S(w) \in (S^\perp)^\perp \\ w-P_S(w) \in S^\perp \end{cases} \Rightarrow \quad (P_S(w), w-P_S(w)) = 0\,. \quad (**)$$

So, we obtain from $(*)$ and $(**)$ that

$$\begin{aligned} \|w-P_S(w)\|_H^2 &= (w-P_S(w), w-P_S(w)) = \underbrace{(w, w-P_S(w))}_{=0 \text{ see } (*)} + \underbrace{P_S((w), w-P_S(w))}_{=0 \text{ see } (**)} \\ &= 0+0 = 0\,. \end{aligned}$$

Therefore, $\|w-P_S(w)\| = 0$ which implies that $w - P_S(w) = 0_H$ and hence,
$w = P_S(w) \in S \;\Rightarrow\; \boxed{w \in S}$. From the twomboxes, it follows that $(S^\perp)^\perp \subset S$.
Conclusion: $(S^\perp)^\perp = S$

(e) Deduce from (b) and (d) that $(S^\perp)^\perp = \overline{S}$.

In fact, we just write

$$(S^\perp)^\perp = \underbrace{(\overline{S}^\perp)^\perp}_{\text{from (b)}} = \underbrace{\overline{S}}_{\text{from (d)}} \qquad \text{(we applied (d) to } \overline{S} \text{ which is a closed subspace)}\,.$$

7. Let $\mathbb{R}^4$ be equipped with the usual dot product. Let

$$F := \{(x, y, z, t) \in \mathbb{R}^4\colon\ x + y + z = 0 \text{ and } x + y + z + t = 0\}\,.$$

Find a basis of F and a basis of $F^{\perp}$.

Basis of F: First of notice that F is a clearly a vector subspace. Now,

$$u = (x, y, z, t) \in F \quad \Leftrightarrow \quad \begin{cases} x + y + z = 0 \\ x + y + z + t = 0 \end{cases} \Leftrightarrow \quad \begin{cases} x + y + z = 0 \\ t = 0 \end{cases} \quad \Leftrightarrow \quad \begin{cases} z = -y - z \\ t = 0 \end{cases}$$

Therefore, we write

$$(x, y, z, t) = (x, y, -x - y, 0) = (x, 0, -x, 0) + (0, y, -y, 0) = x(1, 0, -1, 0) + y(0, 1, -1, 0)\,.$$

Hence,

$$F \subset \mathrm{Span}((1, 0, -1, 0),\ (0, 1, -1, 0)) \subset F \quad \Rightarrow \quad F = \mathrm{Span}((1, 0, -1, 0),\ (0, 1, -1, 0))\,.$$

Since the system $\{(1, 0, -1, 0),\ (0, 1, -1, 0))\}$ is linearly independent, it follows that $\{(1, 0, -1, 0),\ (0, 1, -1, 0))\}$ is a basis of F.

Basis of $F^{\perp}$:

$$v = (x, y, z, t) \in F^{\perp} \quad \Leftrightarrow \quad \begin{cases} ((x, y, z, t), (1, 0, -1, 0)) = 0 \\ ((x, y, z, t), (0, 1, -1, 0)) = 0 \end{cases} \quad \Leftrightarrow \quad \begin{cases} x - z = 0 \\ y - z = 0 \end{cases}$$

Hence,

$$(x, y, z, t) = (z, z, z, t) = z(1, 1, 1, 0) + t(0, 0, 0, 1)\,.$$

Hence, $F^{\perp} = \mathrm{Span}((1, 1, 1, 0),\ (0, 0, 0, 1))$.

Now, notice that the system $\{(1, 1, 1, 0),\ (0, 0, 0, 1))$ is linearly independent.
So, $\{(1, 1, 1, 0),\ (0, 0, 0, 1)\}$ is a basis of $F^{\perp}$.